

Manual for Gait Kinematics Prediction Toolbox and KIST Human Gait Pattern Data

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1 Download and Quick Start

You can download ‘Gait Kinematics Prediction Toolbox’ and ‘KIST Human Gait Pattern Data’ from the below links.

- http://www.me.utexas.edu/~reneu/software/Gait/Gait_Kinematics_Prediction_V1.01_Release.zip
- https://sourceforge.net/projects/gaitkinematicsprediction/files/Gait_Kinematics_Prediction_V1.01_Release.zip/download

After the download, you can quickly see and understand how to use ‘Gait Kinematics Prediction Toolbox’ by simply running the script, `./demo_Gait_Pred.m` in Matlab or Octave. It will produce several figures showing a test subject’s gait kinematics from the subject’s 14 body parameters based on the Gaussian Process Regression (GPR) algorithm and ‘KIST Human Gait Pattern Data.’

2 Introduction

The toolbox and database provides a function mapping from an arbitrary subject’s body parameters to the gait kinematics pattern. The user can predict an arbitrary subject’s gait kinematics by only inputting the subject’s 14 body parameters which are easy to measure. The prediction is performed based on the GPR algorithm (Rasmussen, 2006) and ‘KIST Human Gait Pattern Data’ containing 108 human subjects’ experiment data. The output is a probability distribution (Gaussian distribution) of the predicted gait kinematics, not a strict value. The mean value of the predicted probability distribution provides the gait kinematics trajectory, and the standard deviation value provides the level of confidence for the prediction. For more information, refer to the main article.

For user’s special applications and for the expansion of the toolbox and database, we also provide a function, `./Gait_Model.m`, which creates a new optimal hyperparameter set. The user can make their own prediction model for the user’s new or expanded database.

For user’s convenience, all the programs were made to be compatible with Matlab 7.x and GNU Octave. If a user needs fast prediction or fast optimization, the computation speed can be enhanced by generating MEX file in Matlab/Octave.

The main toolbox ‘Gait Kinematics Prediction Toolbox’ is using another toolbox ‘Gaussian Process Regression and Classification Toolbox’ (Rasmussen, 2011) for GPR algorithm, and the more detail description is in <http://gaussianprocess.org/gpml/code>.

For more detail information associated with the theories and the prediction procedures, refer to the main article, or contact Youngmok Yun (yunyoungmok@gmail.com).

3 KIST Human Gait Pattern Data

‘Kist Human Gait Pattern Data’ has the human experiment data containing their body parameters and gait kinematics patterns. The data were post-processed for ‘Gait Kinematics Prediction Toolbox,’ and its detail procedure is explained in the main article.

- **./database/Data_x.mat** has a matrix containing 108 human subjects’ 14 body parameters. Each row includes one subject’s body parameters, and the index of column has the following sequence: 1. Age, 2. Height (cm), 3. Mass (kg), 4. Gender (0:f 1:m), 5. Thigh Length (mm), 6. Calf Length (mm), 7. Bi-trochanteric Width (mm), 8. Bi-iliac Width (mm), 9. ASIS Breath (mm), 10. Knee Diameter (mm), 11. Foot Length (mm), 12. Malleolus Height (mm), 13. Malleolus Width (mm), 14. Foot Breath (mm).
- **./database/Data_y.mat** contains the gait period and 14 joint motions of the 108 subjects. The gait period is stored in the matrix, *Data_y_period*. 14 joint motions are stored in the matrix, *Data_y*, and its column index has the following sequence: 1. X-Pelvis (mm), 2. Y-Pelvis (mm), 3. Z-Pelvis (mm), 4. Pelvis Yaw (deg), 5. Right Hip Roll (deg), 6. Right Hip Pitch (deg), 7. Right Hip Yaw (deg), 8. Right Knee Pitch (deg), 9. Right Ankle Pitch (deg), 10. Left Hip Roll (deg), 11. Left Hip Pitch (deg), 12. Left Hip Yaw (deg), 13. Left Knee Pitch (deg), 14. Left Ankle Pitch (deg)

For more information associated with the data collection and post-process, refer to the main article.

4 Gait Kinematics Prediction Toolbox

‘Gait Kinematics Prediction Toolbox,’ is written based on another toolbox ‘Gaussian Process Regression and Classification Toolbox,’ and executed based on the database ‘KIST Human Gait Pattern Data.’

‘Gait Kinematics Prediction Toolbox’ contains the below directories and files.

- **./hyp/*** contains the hyperparameter sets which are optimized for the given database, ‘KIST Human Gait Pattern Data.’ These files are loaded when `./Gait_Pred.m` is executed. `./Gait_Model.m` can make a new optimal hyperparameter set for a new database.
- **./GP/gp01pred.m** includes covariance function for the prediction of gait pattern. Its detail is explained in the main article.
- **./Gait_Pred.m**, predicts an arbitrary subject’s gait pattern from his/her body parameters. Its input parameters are ‘body parameter,’ ‘directory path for hyperparameter set,’ and ‘directory path for database.’ The output of the function is the mean value and the standard deviation of the predicted gait pattern. In addition, figures showing the predicted mean, the predicted standard deviation, the gait kinematics trajectories in database, and the average of them are saved with ‘png’ image format.
- **./Gait_Model.m** makes a new optimal hyperparameter set for a newly given database. Its input parameter is ‘directory path for database.’ There is no function output, but it saves the optimal hyperparameter set as files (e.g., `hyp_op_x.mat`).

Warning: The optimization may take longer time than one day for more than 100 subjects data.

- **./demo_Gait_Pred.m** demonstrates the use of `./Gait_Pred.m` function. By just simply running this demo program, the user can understand how to use `./Gait_Pred.m` function. Here in the demo script, we provides three different body parameter sets obtained from middle, tall, and short subjects and one unrealistic body parameter set. Also we provide ‘KIST Human

Gait Pattern Data’ and the hyperparameter set optimized for the database. This demo program is using the database and the hyperparameter set, but the user can select their own database or hyperparameter set by changing the directory path of them.

- `./demo_Gait_Model.m` demonstrates the use of the function `./Gait_Model.m`. This script finds the optimal hyperparameter set for a given database. Because it takes long time to find an optimum solution, the user needs to wait for long time. The optimization speed can be accelerated by using MEX function supported by Matlab/Octave.
- `./GP/*` contains ‘Gaussian Process Regression and Classification Toolbox.’ For more information, refer (Rasmussen, 2011).

5 Copyright

5.1 KIST Human Gait Pattern Data

All copyright associated with ‘KIST Human Gait Pattern Data’ is reserved, and the Center of Bionics at Korea Institute of Science and Technology has the right. This database can be used **only for ‘Gait Kinematics Prediction Toolbox.’** All redistribution and uses for other purposes in source and binary forms, with or without modification, are **not permitted**.

If someone wants to use the database for other purposes, please contact Youngmok Yun (yunyoungmok@gmail.com) or Changhwan Kim (ckim@kist.re.kr).

5.2 Gait Kinematics Prediction Toolbox

Gait Kinematics Prediction Toolbox version 1.0 for GNU Octave and Matlab 7.x

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5.3 Gaussian Process Regression and Classification Toolbox

Gait Kinematics Prediction Toolbox version 1.0 for GNU Octave and Matlab 7.x

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The code and associated documentation is available from <http://gaussianprocess.org/gpml/code>.

6 Version update

1. v1.0

- First release to the reviewers of the Journal of Biomechanics.

2. v1.01

- Several minor bugs are fixed.
- A better hyperparameter set is selected.
- Body parameters of an unrealistic subject are added in the demo prediction file.
- Demo results (figures) are added in the toolbox.

References

Rasmussen, C. E., 2006. Gaussian processes for machine learning.

Rasmussen, C. E., 2011. The gaussian processes web site.

URL <http://gaussianprocess.org>